PEDRAM MOHAMMADI

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PROFESSIONAL SUMMARY

- Innovative and results-driven image and video processing specialist with deep experience translating perceptual principles into hardware- and production-ready solutions. Expertise includes SDR/HDR tone mapping, color processing, encoder optimization, and deep learning-based algorithms. Proven track record of patented innovations, research-to-production delivery, and open-source projects. Proficient in Python, MATLAB, C/C++, OpenCV, and FFmpeg. Comfortable leading cross-functional engineering work and shipping algorithms that measurably improve performance.

EDUCATION

Ph.D. in Electrical and Computer Engineering *2015 - 2020*
University of British Columbia, Canada

- Thesis: High Dynamic Range (HDR) video processing
- Awards: Full graduate scholarship, International Tuition Award

M.A.Sc. in Electrical Engineering *2012 - 2014*
Ferdowsi University of Mashhad, Iran

- Thesis: Subjective and Objective Image and Video Quality Assessment
- Published a widely cited survey paper (300+ citations) on image/video quality assessment

PROFESSIONAL EXPERIENCE

Senior Video Quality Engineer *2021 - Present*
NETINT Technologies Inc., Canada

- Deep learning based video quality enhancement (open source project)
  - Designed and implemented a conditional autoencoder with a learned quantization bottleneck that takes a reference frame and generates the reconstructed frame emulating per-frame encoder distortion, quantization artifacts, without running a full encoder [GitHub].
  - Designed and implemented a learning-based algorithm that predicts the VMAF score between reference and distorted frames using a Siamese 3D-CNN architecture [GitHub].
  - Built a scalable PyTorch pipeline with automated video pre-processing, frame-level quality metric computation, and balanced dataset generation across quality ranges.
  - Implemented advanced training strategies including perceptual loss, auxiliary supervision, and multi-loss scheduling; currently finalizing large-scale training and validation.
- Designed and implemented advanced algorithms to maximize perceptual visual quality for streaming and real-time applications, achieving consistent improvements of up to 15% in objective quality metrics (PSNR, SSIM, VMAF).
- Designed and optimized SDR$\longleftrightarrow$HDR tone mapping suite, outperforming state-of-the-art by 5 - 8% in objective quality metrics, integrated directly into hardware.
- Led end-to-end pipeline tuning and validation across VPU blocks, identifying quality bottlenecks and coordinating targeted improvements with cross-functional teams resulting in an average 4% increase in visual quality (PSNR, SSIM, VMAF) across multiple processing stages.

Scientific R&D Consultant *2020 - 2021*
KPMG LLP, Canada

- Evaluated clients' advanced R&D efforts in algorithm design, image/video processing, and related technologies to determine CRA R&D tax relief eligibility
- Helped clients maximize R&D tax credits, achieving 20 - 30% increases (totaling $500k - $2M), enabling continued investment in innovation
- Managed concurrent client engagements under tight deadlines, developing technical reports and justifications aligned with both financial and scientific rigor

Application Developer *2018 - 2019*
TELUS Communications Inc., Canada

- Transitioned PhD research into a real-time SDR-to-HDR video conversion system used in content delivery pipeline for 1080p broadcast.
- Collaborated with the CTO, directors, and engineers to define visual quality goals and meet real-time constraints, verified through objective and subjective testing.
- Successful technology transfer and product deployment with impact on subscriber-facing video quality.

Research Scientist *2015 - 2020*
The University of British Columbia, Canada

- Conducted PhD research on perceptual SDR-to-HDR video enhancement, modeling contrast and brightness trade-offs and color fidelity in the perceptual domain.
- Developed adaptive segmentation, tone mapping, color adjustment, and flicker mitigation algorithms using visual system models and information theory.
- We conducted extensive subjective studies and applied objective quality metrics (HDR-VDP, PU-SSIM) to validate 70 - 80% improvements over the state of the art.
- The work led to multiple IEEE publications and a patented conversion pipeline that is now licensed in industry.

PATENTS

- **P. Mohammadi**, J. Li, E. Andrade Neto, and J. Lou "Methods and Apparatus for Video Rate Control," Patent filed, filing date: January 2024.
- **P. Mohammadi**, E. Andrade Neto, and J. Lou "Image processing methods, devices, electronic devices and storage media," China Patent Office, Application No. CN116167950B, Publication date: April 2023.
- P. Nasiopoulos, S. Ploumis, M. T. Pourazad, R. Boitar, and **P. Mohammadi**, "Methods and Apparatuses for Tone Mapping and Inverse Tone Mapping," U.S. Patent, Application No. US11100888B2, Publication date: January 2019.

PUBLICATIONS

For a complete list of my publications, please visit my Google Scholar page

WORK AUTHORIZATION

Currently holding a Canadian citizenship